A Simple Accessibility Theory of Parity Circuit Competition

1 Setup

Consider a pure-ReLU residual block. Suppose a neuron combines p orthogonal parent Walsh modes

$$q_1, \dots, q_p, \qquad \|q_i\| = Q,$$

with balanced signs. If v is the neuron's weight vector, the preactivation amplitude of each parent is

$$\gamma = \frac{\|v\|Q}{\sqrt{p}}.$$

For a balanced p-parent ReLU gate, the coefficient of the product Walsh mode is

$$c_p = \kappa_p \gamma,$$

with

$$|\kappa_2| = \frac{1}{2}, \qquad |\kappa_4| = \frac{1}{4}, \qquad |\kappa_8| = \frac{5}{32}.$$

Define the accessibility efficiency

$$R \equiv \frac{|c_p|}{\|v\|} = \frac{|\kappa_p|Q}{\sqrt{p}}.$$

Thus R measures how much of the desired child mode is produced per unit hidden-weight norm. The simple theory predicts that, when several routes to the same target mode are simultaneously available, the route with larger R is more accessible.

2 $2 + 2 \to 4$ versus $1 + 1 + 1 + 1 \to 4$

Let Q_1 be the norm of each relevant singleton mode and Q_2 the norm of each relevant degree-2 mode.

For the direct route $1 + 1 + 1 + 1 \to 4$, we have $p = 4$, $Q = Q_1$, and $|\kappa_4| = 1/4$:

$$R_{1^4} = \frac{(1/4)Q_1}{\sqrt{4}} = \frac{Q_1}{8}.$$

For the compositional route $2 + 2 \to 4$, we have $p = 2$, $Q = Q_2$, and $|\kappa_2| = 1/2$:

$$R_{2+2} = \frac{(1/2)Q_2}{\sqrt{2}} = \frac{Q_2}{2\sqrt{2}}.$$

The two routes are equally efficient when

$$\frac{Q_2}{2\sqrt{2}} = \frac{Q_1}{8},$$

hence

$$\boxed{\frac{Q_2}{Q_1} = \frac{1}{\sqrt{8}} \approx 0.3536.}$$

Therefore

$$\frac{Q_2}{Q_1} > \frac{1}{\sqrt{8}}$$

predicts that $2 + 2 \to 4$ is more accessible, while below this ratio the direct $1^4 \to 4$ route is more accessible.

3 $4 + 4 \to 8$ versus $1 + \cdots + 1 \to 8$

Let Q_4 be the norm of each relevant degree-4 parent mode.

For the direct route $1 + \cdots + 1 \to 8$, we have $p = 8$, $Q = Q_1$, and $|\kappa_8| = 5/32$:

$$R_{1^8} = \frac{5}{32}\frac{Q_1}{\sqrt{8}}.$$

For $4 + 4 \to 8$, we have $p = 2$, $Q = Q_4$, and $|\kappa_2| = 1/2$:

$$R_{4+4} = \frac{Q_4}{2\sqrt{2}}.$$

Equating the two,

$$\frac{Q_4}{2\sqrt{2}} = \frac{5Q_1}{32\sqrt{8}}.$$

Since $\sqrt{8} = 2\sqrt{2}$,

$$\boxed{\frac{Q_4}{Q_1} = \frac{5}{32} = 0.15625.}$$

Therefore

$$\frac{Q_4}{Q_1} > 0.15625$$

predicts that $4 + 4 \to 8$ is more accessible, while below this ratio the direct $1^8 \to 8$ route is more accessible.

4 Experimental method

The theory can be tested while separating circuit accessibility from the dynamics by which the parent features are learned.

1. Train a first ReLU residual block to produce the required parent parity modes. For the degree-4 experiment, train degree-2 parents; for the degree-8 experiment, train degree-4 parents.
2. Freeze the embedding and the trained first block.

3. Rescale the frozen block by
$$V^{(1)} \mapsto \alpha V^{(1)}.$$
By ReLU homogeneity,
$$\mathrm{ReLU}(\alpha z) = \alpha\,\mathrm{ReLU}(z),$$
so this changes the strength of the learned parent modes without changing the learned circuit itself.

4. For every α, directly measure the Walsh-mode norms in the frozen residual stream, in particular
$$\frac{Q_2}{Q_1} \quad \text{or} \quad \frac{Q_4}{Q_1}.$$
These measured ratios, rather than α itself, are the theoretical control variables.

5. Add a fresh ReLU block and readout and train only them on the higher-degree target.

6. Determine which circuit forms from the Walsh decomposition of the new block's preactivations. A $2 + 2 \to 4$ neuron should be dominated by its two degree-2 parents; a $4 + 4 \to 8$ neuron by its two degree-4 parents; a direct route should be dominated by the relevant singleton modes.

The key predictions are
$$\boxed{\frac{Q_2}{Q_1} = \frac{1}{\sqrt{8}}}$$
for the degree-4 experiment and
$$\boxed{\frac{Q_4}{Q_1} = \frac{5}{32}}$$
for the degree-8 experiment.

These predictions assume pure ReLU, approximately equal-norm and orthogonal parent modes, and balanced gates.